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MILITARY SPECIFICATION

BORING AND MILLING MACHINES, PRECISION, HORIZONTAL SPINDLE, TABLE TYPE, NUMERICAL CONTROL (NC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers numerically controlled, horizontal spindle, table type, boring, milling, drilling machines for performing machining operations to precision tolerances with equipment and accessories specified herein.

1.2 Classification. This specification covers machines of the following sizes. The sizes shall be furnished as specified (see 6.2.1).

Size 3 - 3-inch spindle diameter
Size 4 - 4-inch spindle diameter
Size 5 - 5-inch spindle diameter
Size 6 - 6-inch spindle diameter

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified,

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the issue of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the non-Government documents which are current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NFPA 79	- Metalworking Machine Tools and Plastics Machinery, Electrical Standard for.
ANSI B5.1	- T-Slots, Their Bolts, Nuts and Tongues.
ANSI B5.18	- Spindle Noses and Tool Shanks for Milling Machines.
ANSI B5.50	- "V" Flange Tool Shanks for Machining Centers with Automatic Tool Changers.
ANSI B11.8	- Drilling, Milling, and Boring Machines - Safety Requirements for the Construction, Care, and Use.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

ELECTRONICS INDUSTRIES ASSOCIATION (EIA) STANDARDS

RS-232	- Interface between Data Terminal Equipment and Data Communication Equipment Employing Serial Binary Data Interchange.
RS-244	- Character Coded for Numerical Machine Tool Control Perforated Tape.
RS-267	- Axis and Motor Nomenclature for Numerically Controlled Machine Tools.
RS-274	- Interchangeable Perforated Tape Variable Block Format for Positioning, Contouring, and Contouring/ Positioning Numerically Controlled Machine.
RS-281	- Construction Standards - Numerical Machine Tool Control.
RS-358	- Subset of American National Standard Code for Information Interchange for Numerical Machine Control Perforated Tape.

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RS-447 - Operational Command and Data Format for Numerically Controlled Machines

RS-449 - General Purpose 37-Position and 9-Position Interface for Data Terminal Equipment and Data Circuit Terminating Equipment Employing Serial Binary Data Interchange.

RS-494 - 32 Bit Binary CL Exchange (BCL) Input Format for Numerically Controlled Machines.

(Application for copies should be addressed to Electronic Industries Association, Engineering Department, 2001 Eye Street, N.W., Washington, DC 20006.)

NATIONAL ELECTRIC MANUFACTURERS ASSOCIATION (NEMA) STANDARDS

MG-1 - Motors and Generators

(Application for copies should be addressed to the National Electrical Manufactures Association, 155 East 44th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

AMERICAN GEAR MANUFACTURER'S ASSOCIATION (AGMS)

AGMA 360.02 - Manual for Machine Tool Gearing

AGMA 390.03 - Gear Classification Materials and Measuring Methods For Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturer's Association, Standards Department, Suite 1000, 1901 North Fort Meyer Drive, Arlington, VA 22209-1695.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54 - Cylindrical Gears For General Engineering and For Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the International Organization for Standardization, c/o American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018.)

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(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheet or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The machine shall be new and one of the manufacturer's current models capable of precision boring, milling and drilling in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations.. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI), conforming to FED-STD-376 shall be used on graduated measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.1.2 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI Systems of

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Measurement. When a dual US and SI system is furnished, both indicators shall have independent zero adjustments and both indicators shall be calibrated in such a manner that the last graduation progresses into and is continuous with the first graduation as the indicator progresses through the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials specified for use on the machine.

3.2.3 Energy efficiency. The machine and its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.5 Safety and health requirements. Construction of the machine shall be in accordance with ANSI B11.8. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA CFR 1910 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. The machine shall be so designed and constructed that under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prescribed by Federal, state, or local statutes in effect at point of installation (see 6.4).

3.2.9 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include

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a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be readily accessible.

3.2.10 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Dials. All rotating dials or handwheels which are used to position machine axes or for tool movement shall be graduated in increments not larger than .001 inch. Dial and handwheel diameters shall be permanently and legibly engraved or etched on a nonglare background with graduation that can be read from the operator's normal position.

3.3.8 Gears. All gears shall be constructed of a material suitable for the intended purpose. All gears shall be machined and heat treated by a process that will impart the hardness, surface finish, and toughness that will enable the gear train to the transmit full-rated torque of the drive

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motor without gear damage, failure, or premature deterioration and wear. All gears and pinions used in the machine or its components shall be properly selected for the intended application and fabricated to either English (US) standards per AGMA 390.03 and AGMA 360.02 or Metric (SI) standards per the International Organization for Standardization, ISO 54.

3.3.9 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The machine shall consist of not less than a base, column, saddle and worktable, machining head, spindle assembly, coolant system, electrical system and numerical control system.

3.4.1 Base. The base shall be a casting of iron or iron alloy or a fabricated steel weldment, ribbed and braced to minimize distortion and deflection. The base shall be intregally constructed with the column or be separately constructed with the column, machined, securely attached, and in either case, be sufficiently rigid to maintain mutual component alignment for assuring accuracy required herein. The base shall have guideways that are proportioned to fully support the saddle and worktable throughout their full ranges of travel. The base shall have means for leveling the machine on a flat surface. Hydraulic or coolant reservoirs may be housed within the base or be supplied as separate external units. Reservoirs shall be supplied with means for indicating fluid levels. All ways shall be hardened to at least Rockwell C-55, and finished to precision tolerances. In the case where replaceable ways are used, the mating surfaces shall be precision finished to assure flatness of the surface of the way after installation.

3.4.2 Column. The column shall be a casting of iron or iron alloy or a fabricated steel weldment of box-like construction, internally ribbed and braced to minimize distortion and deflection under normal loads. The column shall be designed to support and accommodate the machining head and have guideways to facilitate vertical movement of the head. The column shall be an integrally cast component of the bed or a separate piece mated and securely fastened to the bed to precise alignment tolerances.

3.4.3 Machining head. The machining head shall be a casting of iron or iron alloy or a fabricated steel weldment designed to house and support the spindle assembly and rotational drive with motor and related components necessary for accomplishing boring, milling, drilling and tapping. The head shall hold the spindle in radial alignment parallel to the work holding surface of the worktable and shall be designed to prevent distortion due to the rotational forces of the drives. All machining head functions shall be numerically controlled. The machining head shall be counterbalanced or have equivalent means of counteracting gravitational forces upon the head. The machining head shall be equipped with an automatic power clamp, or servo lock interlocked with the Y-axis drive to prevent applying power to a clamped or locked mechanism. The machine shall have positive stops at the

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limits of travel in each axis to prevent overrun. The spindle shall be precision fitted in the machining head and mounted in anti-friction roller bearings. The construction shall provide for constant preload or adjustment of spindle bearings to maintain accuracy.

3.4.4 Saddle and table. The saddle and table shall move on horizontal guideways for positioning and machining operations. The saddle and table movements shall be power operated in each mode of operation. The slide drive mechanisms shall be independent servo motors with precision recirculating ball screw and nut. Slide motions shall include retracting and repositioning the table on the W axis, and when included in the design, feed movement along the W axis. Each slide shall be equipped with automatic power clamping or servo lock, interlocked with the drive controls to prevent the application of power to a clamped or locked mechanism. All slides shall be equipped with means for preventing chips or other foreign matter from entering between the mating surfaces. Means shall be provided to adjust for wear and to maintain alignment of sliding units. T-slots shall conform to ANSI B5.1 and be spaced for readily clamping work or fixtures. The table and saddle shall be equipped to drain and return coolant to a reservoir. The coolant recovery system shall be filtered to prevent the entrance of chips into the system. The filteration system shall be accessible for either cleaning or replacing, or both. Means shall be provided for indicating coolant reservoir fluid level which will not necessitate the removal of covers or guards.

3.4.5 Spindle assembly. The spindle shall be of alloyed steel hardened and ground to a smooth finish. The spindle shall be held in axial and radial alignment by ball or roller bearings. The spindle shall be mounted in the machining head with its bearings designed and arranged to compensate for thermal expansion. The spindle shall have positive means for accepting, retaining and releasing tools and tool holders inserted by the operator or an automatic tool changer, when specified (see 6.2.1). Unless otherwise specified (see 6.2.1), spindle characteristics (nose, bore) shall conform to the provisions of ANSI B5.18. When an automatic tool changer is used spindle nose characteristics shall conform to ANSI B5.50. The machine shall have overload protection which automatically stops the movement of the slides and spindle rotation should it stall from overload. Spindle rotation shall be automatically stopped for tool exchange in both the manual and automatic modes.

3.4.6 Guideways. The ways on which the saddle, table and machine-head slide for support and guidance shall be constructed to maintain alignment under heavy cutting load to minimize wear and friction. Means for reducing friction such as use of anti-friction materials, airlift devices, or hydrostatic ways shall be of proven design and shall be the manufacturer's standard construction, suitable for maintaining precision performance of the machine to the specified tolerances. All guideway surfaces shall have a surface hardness of not less than Rockwell C-55.

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3.4.7 Tool changer. When specified (see 6.2.1), the machine shall have an automatic tool changer with a magazine capable of storing not less than the tool capacity required for the style machine specified in the contract or order. The magazine shall be of the closed loop design or any other recirculating configuration with the capability of presenting a preprogrammed tool to the tool changer for automatic exchange with tool held in the spindle. Tool change shall be accomplished by means of a power actuated mechanism which shall exchange the tool in the spindle with one from the magazine. The magazine shall permit convenient loading and unloading of tools, tool holders,and adapter during operational set-up. The tool changer shall be arranged for random tool selection which shall include the capability of reusing a tool that has been returned to the magazine after a program operation.

3.4.8 Coolant system. The coolant system shall include a sump or reservoir, a power driven pump and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a baffle or a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant.

3.4.9 Hydraulic system. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to insure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level. When specified (see 6.2.1), a noise control enclosure shall be provided.

3.4.10 Electric system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA 79. Each machine shall draw all of its electrical power from a single 230/460-volt, 3-phase, 60-Hz circuit. Each machine shall have a fused safety disconnect switch or circuit breaker. The machine shall be initially wired for operation on 460 volts. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA 79 for the disconnect fuse rating.

3.4.11 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a dripproof enclosure, and shall comply with the applicable provisions of NEMA MG-1.

3.4.11.1 Solid state components. Solid state design shall be used throughout for electronic components.

3.4.12 Numerical control. The machine shall have a fully automatic solid state soft-wired, integrated circuit type numerical control. The machine

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control unit (MCU) shall provide automatic control of machine functions, operating modes, linear axes movement, spindle operation, tool change, and other part program directed functions for machines of the type specified herein. Controlled axes shall be identified in accordance with EIA Standard RS-267. The control shall have linear and circular interpolation and be capable of simultaneous control of linear axes motion as specified (see 6.2.1). Control features shall include a programmable interface, buffer storage, fixed cycles, part program storage, part program and buffer edit specified herein. The control unit shall comply with EIA-Standard RS-281.

3.4.12.1.1 Rotary table, B axis. When specified (see 6.2.1), the control system shall also be configured for a B axis and the machine shall be furnished with a rotary table. The B axis control shall direct rotation of the rotary table to any of the program selectable (degree) positions specified (see 6.2.1). The rotary table shall be of the contouring type or positioning type as specified (see 6.2.1). The rotary table shall be either an integral component of the machine or an attachment to the worktable. The top of the rotary table shall have T-slots for workpiece attachment. When specified (see 6.2.1), the rotary table shall be constructed so the workpiece shall be rotated about the X axis.

3.4.12.2 Data input.

3.4.12.2.1 Punched tape data input. Unless otherwise specified (see 6.2.1), the input media for the control shall be 1-inch, 8-channel punched tape. The data input format and codes shall comply with EIA Standards RS-274, RS-358 and RS-447 or when specified (see 6.2.1), RS-244 shall apply in lieu of RS-358. The control shall be equipped with a photoelectric tape reader capable of holding, feeding and reading tape prepared in the format specified at a rate not less than 150 characters per second. Unless otherwise specified (see 6.2.1), the tape reader shall be supplied with standard feed and take-up spools 5-1/4 inches in diameter. The tape reader shall be housed in a dust-free area of the numerical control cabinet and shall be easily accessible for cleaning and tape handling. The control shall have a tape error detection capbility to continuously check for the following errors:

a. Odd parity per EIA RS-244.
b. Even parity per EIA RS-358.
c. Misaligned sprocket holes.
d. No sprocket holes.

Any of the above errors shall halt machine functions and illuminate an error light or a message on the data display screen to indicate the fault condition.

3.4.12.2.2 Flexible diskette data input. When 32-Bit Binary Cutter Location (BCL) Exchange Input for Numerically Controlled Machines is specified (see 6.2.1), the input media shall be an 8-inch flexible diskette recorded

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in accordance with 3.4.12.2.2.1. The input data format shall comply with EIA Standard RS-494. Functions 10.1, 10.2 and 10.3 of section 10, EIA-RS-494 shall apply. Manufacturers providing functions or features that cannot be addressed by command codes currently covered by EIA-RS-494 shall apply to and be accepted by the Electronic Industries Association, IE-Committee on Numerical Control Systems and Equipment for an extension of the standard codes to cover the new features or functions. The control shall be equipped with an 8-inch flexible diskette reader. The reader shall have the capability to perform a cyclic redundancy check of the input data and be capable of reading not less than 15,000 characters per second.

3.4.12.2.2.1 Flexible diskette format. When an 8-inch flexible disk is required as the input media to the numerical control, it shall be single-density recording on one side using a soft-sectored format. The disk shall be divided into 77 distinct locations on the single-side disk. Tracks shall be assigned numbers from 0 through 76. Track 0 shall be nearest to the edge of the disk and track 76 nearest the center. Tracks 0, 74, 75, and 76 shall not be used. There is an index hole in the jacket and the disk; the hole shall be used for timing functions when a beam of light passes through it. Recording shall be accomplished by rotating the disk inside its jacket, exposing the disk at the read or write head-slot cut in the jacket. Data shall be recorded only on the side of the disk that is exposed to the read or write head. Each track shall be divided into 26-addressable sectors which contain 120 bits. All sectors shall be identified by the sector address number with track 0, sector 1, designated as 0. The first part program shall be located at sector 1E hexidecimal corresponding to track 1, sector 5. Where two-byte numbers are referenced, the high order byte shall be recorded first and the low order last. The data stored in a sector is called a record. Track 1, sectors 1 through 4 shall contain the program directory of up to 16-part program names in ASCII. Each file shall be packed on the disk in the same order as its directory entry. Unused directory entries shall point to the next available sector after the last file and contain the 30-character ASCII space characters. The last sector of each file shall be padded with end of file (EOF) records (80000000). Diskette addressing shall comply with the requirements of table IV.

3.4.12.2.3 Manual data input. A means for manual data input (MDI) shall be furnished to permit manual entry of alphabetical, numerical, and special characters necessary for control of all programmable machine functions. The manually input data shall be immediately displayed on the data display to permit data verification. The MDI shall be capable of entering a block of data into memory upon command. When specified (see 6.2.1), the control shall be conversationally programmable in the manual data input mode using plain English command data entered through a prompting or menu driven control routine.

3.4.12.3 Data display. A cathode ray tube (CRT) capable of displaying at least 250 characters simultaneously shall be provided. Unless otherwise specified (see 6.2.1), the display shall, as a minimum, display the following data:

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a. Four digit block sequence number.
b. Active block of stored data.
c. Actual position of each axis.
d. Feedrate and override condition.
e. Control and system error message.

3.4.12.4 Control system modes of operation. The operational modes of the NC control shall include manual, automatic, single block, MDI, sequence number search, and optional stop.

3.4.12.5 Departure control. The maximum departure permitted by a single command in either incremental or absolute mode shall be not less than plus or minus 99.9999 inches or 999.999 mm. The control shall be capable of automatically controlling the acceleration and deceleration of the slides. The resolution of slide movement in either direction shall not be more than 0.0001 inch or 0.001 mm.

3.4.12.6 Absolute/incremental input. The control shall have absolute/increment input capability. This mode shall be selected by a preparatory G-code which may be switched during the tape program. Not required if BCL input is specified (see 6.2.1).

3.4.12.7 Automatic tape code recognition. The control shall have the capability to automatically recognize and read tape punched with either EIA RS-244 or RS-358 (ASCII) character codes. Not required if BCL data input is specified (see 6.2.1).

3.4.12.8 Switchable inch/metric input. The control system shall provide switchable inch/metric input capability.

3.4.12.9 Part program storage. Unless otherwise specified (see 6.2.1), part program storage shall be in either solid-state or bubble-type memory. The MCU shall be capable of storing in memory the content of not less than 250 feet of postprocessed part program tape data unless otherwise specified (see 6.2.1). Not applicable if BCL data input is specified (see 6.2.1).

3.4.12.10 Standby battery power. Unless otherwise specified (see 6.2.1), standby battery power shall be provided for controls having volatile memory to hold the memory provided at full load for a period of not less than 72 hours when the primary power source fails.

3.4.12.11 Part program edit. A program edit capability shall be provided to allow block insertion, deletion, modification of any part program in the part program memory. Data related to part program edit shall be input by the manual data input provided on the control panel. Edited program data entered into program memory shall be automatically acted upon in the sequence designated, whenever the part program is cycled.

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3.4.12.12 Feed rate override. A feed rate override control shall be provided to allow manual adjustment of the programmed feed rate from 0 to not less than 120 percent of the programmed rate in either continuous adjustment or in steps of not more than 10 percent. The feed rate override control shall have a position that allows return of feed rate to that programmed.

3.4.12.13 Spindle speed override. A spindle speed override shall be provided to allow manual adjustment of spindle speed in either continuous adjustment or in steps of not greater than 10 percent each. The spindle speed override control shall control not less than the range of override percentages between 60 and 120 percent of programmed speed. The spindle speed override control shall have a position that allows return of speed to that programmed.

3.4.12.14 Sequence number search. The control shall have the capability to permit the operator to search for any desired sequence number.

3.4.12.15 Mirror image. The machine shall be capable of producing either right- or left-hand parts from the same tape data through the inversion of X and Y axes. Inversion of each axis shall be by keyboard or switch entry. Mirror image shall not affect the directional sense of manual controls. Reverse settings shall be displayed on the data display.

3.4.12.16 Preparatory functions (G-codes). The preparatory function codes of the control shall be in accordance with EIA Standard RS-274. If the BCL data input is specified, codes in accordance with EIA Standard RS-494 shall be applied.

3.4.12.17 Block delete. The control shall have a block delete feature to permit by-passing of blocks of programmed data.

3.4.12.18 Buffer storage. The control shall have buffer storage for transferring command data from the tape reader to internal storage without delaying the next incoming command, and without interrupting the machine functions. The internal storage shall delay or store multiple blocks of data until required by the controlling devices.

3.4.12.19 Block-by-block read. The control shall have a block-by-block read function to provide for reading information one block at a time.

3.4.12.20 Tool length offsets. The control shall have a tool length offset feature to permit adjustments to the programmed tool length by means of manual data input. The tool length offset feature shall have a range of not less than the following:

a. Inch system $\pm$ 0.0002" to $\pm$ 1.0000"
b. Metric system $\pm$ 0.002 mm to $\pm$ 25.400 mm.

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3.4.12.21 Cutter compensation. Cutter compensation shall be provided to compensate for cutter diameters differing from those utilized in programming. The cutter compensation shall have a range of not less than the following:

a. Inch system ± 0.0002" to ± 1.0000".
b. Metric system ± 0.002 mm to ± 25.400 mm.

3.4.12.22 Fixture compensation. The fixture compensation feature shall be provided to permit compensation for fixture position errors.

3.4.12.23 Reversal error compensation. Reversal error compensation shall be provided for each axis of motion to compensate for drive gear backlash when slide reversal occurs. A means shall be provided to input new compensation values when required.

3.4.12.24 Axis calibration. Error compensation shall be provided to correct repeatable errors determined to exist due to leadscrew and feedback device inaccuracies.

3.4.12.25 Additional features. Unless otherwise specified (see 6.2.1), the following additional numerical control features shall be provided:

a. Four quadrant programming.
b. Incremental and continuous jog for each axis.
c. Leading zero suppression or decimal point programming.
d. Inches per minute (IPM)/Millimeters per minute (MMPM), programming in all modes of operation
e. Tape edit keylock.

3.4.12.26 Operator control panel. Unless otherwise specified (see 6.2.1), the control unit shall have a control panel which provides at least the following control functions: Emergency Stop; Cycle Start; Optional Stop; Mode Selection; Feed Hold; Feedrate Override; Spindle Speed Override; Jog; Jog Direction; and EOF Stop.

3.4.12.26.1 Pendant control. When specified (see 6.2.1), a pendant control shall be furnished as an operator control panel. The pendant control shall be connected to the machine to enable the operator to control machine functions while in a close observation position. Unless otherwise specified (see 6.2.1), the pendant control shall be capable of the same functions as stated for the operator control panel in paragraph 3.4.12.27.

3.4.12.27 Reference zero. The control unit shall have a reference zero to move each axis to a reference limit switch to establish synchronization and reset all registers to zero.

3.4.12.28 Set zero. The control unit shall have a "set zero" feature to permit the specified axis register to be reset to zero establishing present axis location at zero.

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3.4.12.29 Program tryout. The control shall have a program tape tryout feature to permit new part programs to be read through the control without machine motions.

3.4.12.30 Operational software. Operation software shall be resident in the control and shall contain all logic necessary to effectively operate the machine tool. The software shall provide for meaningful diagnostics relating to the utilization of the controls, machine tool, part program, and operator's actions.

3.4.12.30.1 Programmable interface. A software program for accommodating the softwired interface between the numerical control and machine shall be provided as part of the executive program or as a separate program in the form of 1-inch, 8-channel punched tape complying with EIA-RS-244 or 8-inch flexible diskette as appropriate for the input media specified (see 6.2.1).

3.4.12.31 Maintenance diagnostics system. Maintenance diagnostics software shall be furnished in either the control as non-volatile memory or the form of 1-inch, 8-channel punched tape complying with EIA-RS-244 or 8-inch flexible disk as appropriate for the input media specified (see 6.2.1) to provide for diagnostic checking of the machine control unit. The diagnostic software shall test, exercise, and display failures, at least to board level.

3.4.12.32 Peripheral equipment interface. Unless otherwise specified (see 6.2.1), the control shall be equipped with an interface complying with RS-232 and RS-449.

3.4.12.33 Axis jog. The control panel or pendant shall have manual controls for accomplishing both single and continuous, low and high speed jog movements of the slides for each controlled axis in both the plus and minus directions.

3.4.12.34 Auxiliary functions. Unless otherwise specified (see 6.2.1), the NC shall respond to the following program commanded auxiliary functions: interpolation, fixed cycles, and dwell. Unless otherwise specified (see 6.2.1), miscellaneous functions shall include program stop, optional program stop, end-of-program, spindle CW, spindle CCW, spindle off, coolant on-off, tool change, end-of-program (rewind), and program re-start. System preparatory and miscellaneous functions shall be coded in accordance with EIA RS-274 and EIA RS-447.

3.4.12.35 Canned cycles. The following canned cycles shall be provided:

a. Drilling.
b. Boring.
c. Tapping.

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3.4.12.35.1 Macro program cycles. Flexible macro program shall be provided as specified (see 6.2.1).

3.4.12.36 Tape punch unit. When specified (see 6.2.1), the system shall have a tape punch unit for producing punched tapes from edited programs of tape being input and from programs stored in data memory. The unit shall have a plug receptacle and circuitry that is compatible with the tape punch output of the system, punching both paper and mylar tapes of dimensions, characteristics and codes in accordance with 3.4.12.2. The punching speed of the unit shall be not less than 75 characters per second.

3.4.12.37 Postprocessor. When specified (see 6.2.1), the system shall be furnished with a postprocessor program(s) to be utilized in off-line computer assisted preparation of part programs. The postprocessor shall be provided for converting specified programming language (see 6.2.1) to the form and format required to operate the machine tool and control. The postprocessor shall be written for use on the type computer specified (see 6.2.1). Documentation sufficient to implement and utilize the postprocessor shall be provided for computer compatibility verification and validation at least 30 days prior to delivery of the machining center.

3.4.12.38 Graphics software. When specified (see 6.2.1), graphics software shall be provided which will allow communication between the specified (see 6.2.1) user graphics system and the machine control unit.

3.4.13 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with the machine.

a. A load meter installed on the machine or MCU console that constantly registers the horsepower load (horsepower units or percent of load) on the spindle rotational drive motor.

b. One set of special wrenches, operating, and repair tools that are normally furnished by the manufacturer.

3.4.14 Optional equipment. The following optional equipment shall be furnished as specified (see 6.2.1).

a. Worklight, conveniently mounted to illuminate the work area, complete with protective shield and on-off switch.

b. An elasped time indicator of the non-reset type, with digital readout and designed to resist the effects of vibration, voltage fluctuation, temperature change, and humidity. The indicator shall be mounted on the machine in a position visible from the operator's station and be connected to record and accumulate machine operating time to not less than 9999.9 hours.

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c. Tools, tool holders, and adapters, as specified.

d. Other optional equipment as specified.

3.4.15 Control circuit voltage. Auxiliary control circuits shall be isolated from the input electrical power supply by a low-voltage transformer having a secondary voltage no greater than 115 volts.

3.5 Sizes and capacities. Unless otherwise specified (see 6.2.1), the sizes and capacities of the boring machine shall meet the requirements of table I as applicable for the machine sizes specified in the contract or order.

3.6 Performance. The machine and numerical control system combination shall be capable of performing as specified herein. In addition, the machine shall be capable of boring, milling, drilling and tapping to accomplish the capacities of table I while meeting the accuracies of table II.

3.7 Alignment tolerances. Each machine shall meet the alignment tolerances specified in tables II, and III.

3.8 Repair parts. Such repair parts as are specified (see 6.2.1) shall be furnished.

3.9 Marking on plates, charts, and decals. All words on plates, charts, and decals shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. When these methods are not practical, the instructions shall be applied directly onto the item by environmentally protected decalcomania transfer, metal wrap-around tag, stencil, silk screen, or by any other method suitable for the use intended which shall be as permanent as the normal life expectancy of the item to which it is applied and capable of withstanding the environmental test and cleaning procedures specified for the item.

3.9.1 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.9.2 Nameplate. A nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's 19 permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

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Nomenclature
Manufacturer's name
Manufacturer's model designation
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.10 Position feedback device. A position feedback device, including all related components, hardware, and electrical equipment, shall be provided for each axis having a readout system. The feedback device shall be of the independent type. The length of the feedback device for each axis specified shall be at least 1-3/4 inches longer than the full axis travel to prevent damage to the feedback device in the event of overtravel. Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device. The device shall provide axis travel feedback over the entire range of axis travel for each specified axis. The feedback device shall be the manufacturer's standard type. The feedback devices shall be guarded to prevent damage to them during operation of the machine.

3.11 Technical data. When technical data (operating manuals, maintenance manuals, parts catalogs, prints, wiring diagrams, lubrication charts, machine alignment and accuracy test results) is required it shall be furnished in accordance with the requirements of the DD Form 1423. All technical data furnished shall be written in the English language.

3.12 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4.).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination or any tests shall be cause for rejection.

4.5 Quality conformance inspection Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, or inspection shall be cause for rejection.

4.6 Examination. The machine shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Alignment tolerance tests. Each machine shall be tested to determine that it complies with the alignment tolerance requirements of 3.7.

4.7.2 Performance tests. Numerical controlled boring machines shall be checked for positioning accuracy and repeatability. Accuracies shall be within the tolerances of 3.7 when evaluated on a three-sigma basis in accordance with the National Machine Tool Builder's Association (NMTBG) Definition and Evaluation of Accuracy and Repeatability of Numerically Controlled Machine Tools, Section I - Linear Definition and Section 2 - Angular Definition. At least 10 random target points shall be used, spaced over the limits of travel, when evaluating the accuracies on each axis of motion. If the system uses bi-directional approach, each target point shall be evaluated from both directions. If necessary, the zero reference point may be shifted to permit checking within the capacity of the instruments. For angular axis test, at least four random angles shall be selected utilizing the 360° rotation of the table top. If bi-directional, the target angle shall be evaluated from both directions.

4.7.3 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.9.

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4.7.4 Operational tests. The macine shall be operated in accordance with the builder's standard operating test procedure for warm-up and run-off checks. The run-off checks shall include tests of all mechanical and elec--trical functions of the machine and controls to determine proper functioning in each mode of operation provided. Machines shall be cycled continuously for a period of not less than four hours to check response to commands for all numerically controlled machine functions and auxiliary functions. The commands shall include positioning of each axis separately and in combination, spindle speed changes, feed rate changes, and checks of manual overrides. Should a malfunction occur, the cycle shall be restarted until a full four-hour cycle is successfully completed. On machines equipped with numerical control units, which have manual data input only, the machine shall be cycled for a period of not less than 1 hour to verify proper machine response to inserted command data. The manual input shall be used to check each mode of operation including slide departure speed and feed selections and auxiliary function.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The boring and milling machines covered by this specification are intended for use in machining parts of various configurations and materials to precision tolerances in toolroom and production operation.

6.2 Ordering data.

6.2.1 Procurements requirements. Purchaser should specify their requirements in the procurement documents, including whether each choice is required or not, by entering an appropriate statement identified to each of the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2).

c. First article approval, if required (see 3.1).

d. System of measurement (see 3.2.1).

e. Measuring and indicating device calibrations, (see 3.2.1.1).

f. Specify dual calibration, if required (see 3.2.1.2).

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g. Specify additional safety and health requirements, if any (see 3.2.5).

h. Painting, if different (see 3.3.5).

i. Specify an automatic tool changer, if required (see 3.4.7).

j. Spindle characteristics, if different, (see 3.4.5).

k. Electromagnetic interference control, if required (see 3.3.9).

l. Specify electrical system conformance requirements, if different (see 3.4.10).

m. Motor enclosure type, if different (see 3.4.11).

n. Specify coolant system configuration, if different, (see 3.4.8).

o. Specify noise control features on hydraulic system, if required (see 3.4.9).

p. Specify EIA-RS-494, Binary Cutter Location (BCL) data, if required (see 3.4.12).

q. Specify the program selectable (degree) positions required (72-360 - 360,000) and whether the B axis is required to function simultaneously or independently of the X and Y axes of motion, if a rotary table is required (see 3.4.12.1.1).

r. Specify EIA-RS-244 in lieu of EIA-RS-358 (see 3.4.12.2.1).

s. Specify type of take-up spools, if not standard (see 3.4.12.2.1).

t. Specify BCL Exchange Input, when required (see 3.4.12.2.2).

u. Specify format of conversational program, if required (see 3.4.12.2.3).

v. Specify display data, if different, (see 3.4.12.3).

w. Specify means of part program storage, if different (see 3.4.12.2.2).

x. Specify part program memory capacity, if different, (see 3.4.12.9).

y. Standby battery power, if required (see 3.4.12.10).

z. Specify additional numerical control features, if different (see 3.4.12.25).

aa. Operator control panel features, if different (see 3.4.12.26).

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bb. Operator control panel features, if different (see 3.4.12.26).

cc. Specify a pendant control, if required, and control features, (see 3.4.12.26.1).

dd. Specify 1-inch punched tape or 8-inch flexible disk as MCU program input and storage media (see 3.4.12.30.1).

ee. Specify type of interface, if different (see 3.4.12.32)

ff. Specify auxiliary and miscellaneous functions if different (see 3.4.12.34).

gg. Macro programming capabilities (see 3.4.12.35.1)

hh. Tape punch unit, if required (see 3.4.12.36).

ii. Postprocessor, specify computer name and model, form and format (see 3.4.12.37).

jj. Graphics software (specify graphics system's name and model, form and format), if required (see 3.4.12.38).

kk. Specify standard equipment, if different (see 3.4.13).

ll. Optional equipment, if required (see 3.4.14).

mm. Specify sizes and capacities (see 3.5).

nn. Specify repair parts to be furnished with the machine, if any (see 3.8).

oo. Lubrication chart, if different (see 3.9.1).

pp. Nameplate, if different (see 3.9.2).

qq. First article inspection, if different (see 4.4).

rr. Quality conformance inspection requirements, if different (see 4.5).

ss. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level required and specification number (see 5.1).

tt. Specify any additional safety and health requirements (see 6.4).

6.2.2 Contract data requirements. Required technical data (operating manuals, parts lists, wiring diagrams, foundation and anchor bolt plans and

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acceptance test reports) should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test, and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants and heat. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazards level of individual machine in the environment) the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional requirements that will integrate the new machine into its future operating environment. The machine shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required, since the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1(d).

6.5 Cross-reference of classification changes. Boring machine classification changes from the previous issue of this specification are as follows:

Delete: Type I - Manual pendant control
Type II - Semi-automatically controlled-manual data input system

6.6 Subject term (keyword) listing.

Binary Cutter Location (BCL)
Control, Computer Numerical
Diskette, Flexible
Machine, Boring
Table Types
Table, Rotary
Tape, Punched

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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TABLE I. Sizes and capacities.

Machine Characteristics	Sizes (Dimensions in inches)			
	Size 3	Size 4	Size 5	Size 6
Spindle bar diameter (nominal)	3	4	5	6
Spindle nose taper, ANS or NMTBA, ISO	No. 40, 40	No. 50,45	No. 50	No. 50,50
Spindle bar travel Z axis	16	20	24	36
Spindlehead vertical travel Y axis	36	36	48	60
Spindle speed range RPM	28 to 1400	27 to 1300	25 to 3000	10 to 2500
Spindle feed range IPR or	0.0013 to 0.167	0.0013 to 0.167	0.0013 to 0.167	0.0013 to 0.167
Z axis - CNC IPM	0.1 to 98	0.1 to 98	0.1 to 298	0.1 to 98
Spindle rapid traverse trade IPM	100	100	300	300
Table size, width by length (see options)	31 by 31	35 by 48	40 by 60	48 by 72
Table or saddle trave (retract) W axis	16	16	18	24
Table travel X axis	48	48	60	72
Feed range, milling, boring IPR or	0.001 to 0.999	0.001 to 0.999	0.001 to 0.999	0.001 to 0.999
IPM	0.5 to 150	0.5 to 36	0.5 to 250	0.6 to 350
Rapid travel rate IPM	150	150	150	350
Main drive motor horsepower,				
2 speed or	15/7-1/2	20/10	20/10	30/15
single speed	15	20	25	25
Optional hp	-	-	-	30 or 40
Options: 1/				
Extended column, vertical travel, Y axis	48,60 or 72	48, 60 or 72	60, 72, 84 or 96	72, 84, 96 or 120
Table size, width by length	35 by 60, 72, 96 or 40 by 60,72, 96 120	35 by 60, 72, 96 or 40 by 60, 72, 96 120	40 by 72, 96, 108, 120 or 48 or 60 by 72, 96, 108 120, 144	48 by 96, 108, 120, 144 or 60 or 72 by 96, 108, 120, 144
Index table size:				
Round - 30", 36", 40", 50", 60" dia.				
Square - 20", 30", 40", 48", 60"				
Rectangular - 25" by 36", 36" by 48" 36" by 60", 48" by 60" 48" by 70"				

1/ NOTE: Machine sizes and capacities shall be not less than the stated requirements of the size ordered nor greater than the requirements of the next larger size shown in above table. When the largest size and capacity machine is ordered the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the minimum or less to the maximum or greater.

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TABLE II. Alignment tolerances.

Characteristics		Tolerances (in.)
Table top flatness, concave		0.00010 per foot
Machine components parallel and square		0.00010 per foot
Table edge and T-slots parallel to axis of motion		0.00015 per foot
Cross slots square with T-slots and parallel with table motion		0.00015 per foot
Spindle head vertical travel square with table top, each plane		0.00010 per foot
Spindle bar runout:	0 inches	0.00020 TIR
	bar extended 12 inches	0.00050 TIR
	bar extended 24 inches	0.00100 TIR
Spindle taper runout	1-1/4 inches from nose	0.00030 TIR
	12 inches from nose	0.00050 TIr
Alignment of spindle and table motion, parallel axes		0.00010 per foot

TABLE III. Positioning accuracies numerical control.

Characteristics		Tolerances (in.)
Absolute position accuracy, each longitudinal axes		
(table, saddle, spindlehead)	To 60 inches	±0.0003
	To 96 inches	±0.0004
	To 144 inches	±0.0005
Repeatability		±0.0001
Resolver accuracy (spindle)	To 36 inches	±0.0015
Repeatability		±0.0005
Rotary position accuracy (Index Table)		
with locating pin		±2 Sec/Arc
without locating pin		± Sec/Arc

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TABLE IV. Diskette addressing.

Sector address	Track number	Sector number	Byte	Use
0-25	0	All	All	Unused
26-29	1	1-4	0-1	File start sector address (2 byte binary)
			2-31	30 character ASCII program to be defined by user
			32-63	Same as 0-31 above for next part program
			64-95	Same as 32-63 above for next part program
			96-127	Same as 32-63 above for next part program
30-51	1	5-26	All	BCL Records for NC part programs
52-1923	2-73	All	All	BCL records for NC part programs
1924-2001	74-76	All	All	Unused

1/ This numbering corresponds to the 3740 physical format.

2/ Where 2-byte numbers are referenced, the high order byte is recorded first and low order last.

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Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Army - AR
Navy - YD
Air Force - 84
DLA - GS

User Activities:
Army - AR
Navy - AS, MC, OS

Preparing activity:
DLA - IP

Project (3411-0063)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-80004C	Boring and Milling Machines, Precision, Horizontal Spindle, Table Type, Numerical Control (NC)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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